

Playwright Automation Test Scripts

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1. File: Autosuggestdropdown.spec.ts

```
//Bootstrapdropdown
//1)static drop down(select tag)
//A static dropdown is simply a dropdown built using the HTML <select> tag`, where the options are already
present in the DOM (they don't load dynamically from an API).
//2)Dynamic drop down/Auto suggest drop down
//A dynamic dropdown is a dropdown NOT built with the <select> tag.
//Its options are usually loaded at runtime (via JavaScript / API) and rendered using tags like
//3)hidden drop down
//A hidden dropdown is a dropdown where the options exist in the DOM but are not visible until some user
action (hover, click, focus).
//Automation fails if you try to click options before they become visible.
//Autosuggest Drop down
import{test,expect,Locator} from "@playwright/test";
test("Auto suggest drop down",async({page})=>{
  await page.goto("https://www.flipkart.com/");
  await page.locator("input[name='q']").fill("smart");
  await page.waitForTimeout(5000);
  //get all suggested options -->ctrl+shift+p on DOM-->emulated focused page
  const options:Locator=page.locator("ul>li");
  const count=await options.count();
  console.log("Number of suggested options:",count);
  await page.waitForTimeout(3000);
  //return the text of particular item
  //console.log("5th option:",await options.nth(5).innerText());
  //printing all the suggested options in the console
  for(let i=0;i<count;i++)
  {
    //console.log("5th option:",await options.nth(i).innerText());
    console.log(await options.nth(i).textContent());
    //console.log(await options.nth(i).allTextContents());
  }
  await page.waitForTimeout(2000);
  //select/click on the smartphone option
  for(let i=0;i<count;i++)
  {
    const text=await options.nth(i).innerText();
    if(text==='smartphone')
    {
      await options.nth(i).click();
      await page.waitForTimeout(2000);
      break;
    }
  }
});
```

2. File: Datepicker.spec.ts

```
//Date pcikers
import { test, expect, Locator, Page } from '@playwright/test';
async function selectDate(targetyear:string,targetmonth:string,targetdate:string,page:Page,isFuture:boolean)
{
  while(true)
  {
    const selectedmonth=await page.locator('.ui-datepicker-month').textContent();
    const selectedyear=await page.locator('.ui-datepicker-year').textContent();

    if(selectedmonth?.trim()===targetmonth && selectedyear?.trim()===targetyear)
    {
      break;
    }
    if(isFuture)
    {
      await page.locator('.ui-datepicker-next').click(); //Future
    }
    else{
      await page.locator('.ui-datepicker-prev').click(); //Past
    }
    //await page.waitForTimeout(2000);
  }
  /*const allDates=await page.locator(".ui-datepicker-calender td").all();
  for(let dates of allDates)
  {
    const dateText= await dates.textContent();
    if(dateText===date)
    {
      await dates.click();
      break;
    }
  }
  */

  const allDates=await page.locator(".ui-datepicker-calendar td").all();
  for(let dates of allDates)
  {
    const dateText= await dates.textContent();
    if(dateText===targetdate)
    {
      await dates.click();
      break;
    }
  }
}
```

```
}  
test("Date picker",async( {page})=>{  
  await page.goto("https://testautomationpractice.blogspot.com/");  
  const dateInput:Locator=page.locator("#datepicker");  
  await expect(dateInput).toBeVisible();  
  //expect(dateInput).toBeVisible();  
  //1)using fill() method  
  /*datepicker.fill("07/02/2026"); //mm/dd/yy  
  await page.waitForTimeout(5000);*/  
  await dateInput.click(); //opens date picker  
  /*//past target date  
  const year='2023';  
  const month='May';  
  const date='22';*/  
  //Future target date  
  const year='2028';  
  const month='July';  
  const date='14';  
  await selectDate(year,month,date,page,true);  
  const expecteddate='07/14/2028';  
  await expect(dateInput).toHaveValue(expecteddate);  
  await page.waitForTimeout(2000);  
});
```

3. File: action.spec.ts

```
//Action
//action is used to perform operations on the web elements
//like click, type, select etc.
//input text action
import {test, expect, Locator} from "@playwright/test";
test('Text input action demo', async ({page}) => {
  await page.goto("https://testautomationpractice.blogspot.com/");
  const textbox:Locator=page.locator('#name');
  await expect(textbox).toBeVisible();
  await expect(textbox).toBeEnabled();
  const maxLength:string | null=await textbox.getAttribute("maxlength"); //returns the max length of the textbox
  //await textbox.fill("Playwright Automation");
  //await expect(textbox).toHaveValue("Playwright Automation");
  expect(maxLength).toBe("15");
  await textbox.fill("Jhon Cannady");
  console.log("text content of Firstname:",await textbox.textContent());
  console.log("value of Firstname:",await textbox.inputValue());//inputValue() returns the value entered in the
  textbox
  const enteredValue:string=await textbox.inputValue();
  expect(enteredValue).toBe("Jhon Cannady");
  await page.waitForTimeout(3000);
});
//Radio button action
//only used to execute only this perticular test
test('Radio button action demo', async ({page}) => {
  await page.goto("https://testautomationpractice.blogspot.com/");
  const maleRadio:Locator=page.locator('#male');
  await expect(maleRadio).toBeVisible();
  await expect(maleRadio).toBeEnabled();
  await maleRadio.check();
  expect(await maleRadio.isChecked()).toBe(true);
  expect(maleRadio).toBeChecked
  await page.waitForTimeout(3000);
  console.log("check for radion button selected",await maleRadio.isChecked());
});
//check box action
test.only('Check box action demo', async ({page}) => {
  await page.goto("https://testautomationpractice.blogspot.com/");
  //using getByLabel specific check box
  const sundaycheckbox:Locator=page.getByLabel('Sunday');
  await sundaycheckbox.check();
  expect(await sundaycheckbox.isChecked()).toBe(true);
  await expect(sundaycheckbox).toBeChecked();
  console.log("Check box is selected:",await sundaycheckbox.isChecked());
  //select all the chek box
```

```

const days:string[]=['Sunday','Monday','Tuesday','Wednesday','Thursday','Friday','Saturday'];
const checkboxes:Locator[]=days.map(index=>page.getByLabel(index));
expect(checkboxes.length).toBe(7);
//make all check box marked
for(const checkbox of checkboxes)
{
    await checkbox.check();
    await expect (checkbox).toBeChecked();
}
await page.waitForTimeout(2000);
//uncheck last 3 checkbox and assert
for(const checkbox of checkboxes.slice(-3))
{
    await checkbox.uncheck();
    await expect(checkbox).not.toBeChecked();
}
await page.waitForTimeout(5000);
//Toggle checkbox:if checked uncheck,if unchecked ,check.
for(const checkbox of checkboxes)
{
    if (await checkbox.isChecked())
    {
        //only if unchecked
        await checkbox.uncheck();
        await expect(checkbox).not.toBeChecked();

    }
    else{
        //only if checked
        await checkbox.check();
        await expect(checkbox).toBeChecked();
    }
}
await page.waitForTimeout(2000);
//random check box select
const indexes=[1,3,6];
for(const i of indexes)
{
    await checkboxes[i].check();
    await expect(checkboxes[i]).toBeChecked();
}
await page.waitForTimeout(2000);
//select check box based on the label
const weekname:string="Friday";
for(const label of days)
{
    if(label.toLowerCase()===weekname.toLowerCase())
    {

```

```
const checkbox=page.getByLabel(label);
checkbox.check();
await expect(checkbox).toBeChecked();
}
}
await page.waitForTimeout(3000);
});
```


4. File: authenticatepopup.spec.ts

```
//Authenticated popup test
// it is used to handle the authenticated popup in the browser
// it is used to handle the authenticated popup in the browser by providing the username and password in the
URL
import { test, expect, Locator, Page, chromium } from '@playwright/test';
test('Authenticated popup demo', async ({ browser }) => {
  // □ EXTRA ADDED HERE (authentication)
  const context = await browser.newContext({
    httpCredentials: {
      username: 'admin',
      password: 'admin'
    }
  });
  const page = await context.newPage();
  // Approach 2: directly pass login along with url
  await page.goto('https://the-internet.herokuapp.com/basic_auth');
  await page.waitForLoadState(); // wait for page loaded completely
  await expect(page.locator('text=Congratulations')).toBeVisible();
  await page.waitForTimeout(5000);
});
```

5. File: autoscroll.spec.ts

```
//Automatic scrolling test
// it is used to handle the automatic scrolling in the browser when the element is not visible in the viewport
import { test, expect } from '@playwright/test';
test('Automatic scrolling demo', async ({ page }) => {
  await page.goto('https://demowebshop.tricentis.com/');
  //footer element locator=auto scroll to the footer element
  const footer:string=await page.locator('.footer-disclaimer').innerText();
  console.log("Footer text: "+footer);
  await page.waitForTimeout(5000);
});
test('scrolling inside dropdown', async ({ page }) => {
  await page.goto('https://testautomationpractice.blogspot.com/');
  await page.locator('#comboBox').click();
  const option =page.locator('#dropdown div:nth-child(100)'); // locator for the 10th option in the dropdown
  console.log("Option captured from dropdown:",await option.innerText());
  await option.click(); // click on the 10th option in the dropdown
  await page.waitForTimeout(3000);
});
test.only('scrolling inside the table', async ({ page }) => {
  await page.goto('https://datatables.net/examples/basic_init/scroll_xy.html');
  const name=await page.locator('tbody tr:nth-child(10) td:nth-child(2)').innerText(); // locator for the 50th row
  and 1st column in the table
  console.log("Name captured from table:",name);
  const email=await page.locator('tbody tr:nth-child(10) td:nth-child(9)').innerText(); // locator for the 50th row
  and 2nd column in the table
  console.log("Email captured from table:",email);
});
```

6. File: bootstrapdatepicker.spec.ts

```
import { test, expect } from '@playwright/test';

test('Booking.com Date Picker Test - Check-in and Check-out', async ({ page }) => {
  await page.goto('https://www.booking.com/');
  // Open calendar
  await page.getByTestId('searchbox-dates-container').click();
  // ===== CHECK-IN DATE =====
  const checkinYear = "2026";
  const checkinMonth = "June";
  const checkinDay = "20";
  // Wait for calendar header to appear
  await page.locator("h3[aria-live='polite']").first().waitFor({ state: 'visible' });
  let attempts = 0;
  while (attempts < 24) {
    const headerText = await page
      .locator("h3[aria-live='polite']")
      .nth(0)
      .textContent();
    if (!headerText) throw new Error('Check-in month header not found');
    const [currentMonth, currentYear] = headerText.split(" ");
    if (currentMonth === checkinMonth && currentYear === checkinYear) {
      break;
    }
    await page.locator('button[aria-label="Next month"]').click();
    attempts++;
  }
  if (attempts === 24) {
    throw new Error('Check-in month/year not reached');
  }
  // Select check-in day
  const checkinDates = await page
    .locator('table.b8fcb0c66a tbody')
    .nth(0)
    .locator('td')
    .all();
  let checkinDateSelected = false;
  for (const date of checkinDates) {
    const dateText = await date.innerText();
    const classes = await date.getAttribute('class');
    if (
      dateText === checkinDay &&
      !classes?.includes('b8fcb0c66a--disabled')
    ) {
      await date.click();
      checkinDateSelected = true;
      break;
    }
  }
}
```

```

    }
  }
  expect(checkinDateSelected).toBeTruthy();
  // ===== CHECK-OUT DATE =====
  const checkoutYear = "2026";
  const checkoutMonth = "July";
  const checkoutDay = "25";
  await page.locator("h3[aria-live='polite']").nth(1).waitFor({ state: 'visible' });
  attempts = 0;
  while (attempts < 24) {
    const headerText = await page
      .locator("h3[aria-live='polite']")
      .nth(1)
      .textContent();
    if (!headerText) throw new Error('Check-out month header not found');
    const [currentMonth, currentYear] = headerText.split(" ");
    if (currentMonth === checkoutMonth && currentYear === checkoutYear) {
      break;
    }
    await page.locator('button[aria-label="Next month"]').click();
    attempts++;
  }
  if (attempts === 24) {
    throw new Error('Check-out month/year not reached');
  }
  // Select check-out day
  const checkoutDates = await page
    .locator('table.b8fcb0c66a tbody')
    .nth(1)
    .locator('td')
    .all();
  let checkoutDateSelected = false;
  for (const date of checkoutDates) {
    const dateText = await date.innerText();
    const classes = await date.getAttribute('class');
    if (
      dateText === checkoutDay &&
      !classes?.includes('b8fcb0c66a--disabled')
    ) {
      await date.click();
      checkoutDateSelected = true;
      break;
    }
  }
  expect(checkoutDateSelected).toBeTruthy();
  await page.waitForTimeout(5000); // optional visual check
});

```


7. File: browsercontext.spec.ts

```
//Browser context test
// it is used to create incognito browser context and perform actions in that context
// it is used to create multiple browser contexts in the same test and perform actions in those contexts
import { test, expect, Locator, Page, chromium } from '@playwright/test';
/*test('Browser context demo', async ({ browser }) => {
chromium.launch(); // launch the browser
const context=await browser.newContext(); // create a new incognito browser context
const page=await context.newPage();
await page.goto('https://testautomationpractice.blogspot.com/');*/
//creating multiple pages in the same browser context
test('Browser context demo', async () => {
const browser=await chromium.launch();
const context=await browser.newContext();
const page1=await context.newPage();
const page2=await context.newPage();
console.log("Number of pages created:",context.pages().length);
await page1.goto('https://playwright.dev/');
await expect(page1).toHaveTitle(/Playwright/);
await page1.waitForTimeout(5000);
await page2.goto('https://www.selenium.dev/');
await expect(page2).toHaveTitle(/Selenium/);
await page2.waitForTimeout(5000);
});
```

8. File: comparingmethod.spec.ts

```
//static web table:
import {test,expect, Locator} from "@playwright/test";
test("comparing methods",async({page})=>{
  await page.goto('https://demowebshop.tricentis.com/');
  const products:Locator=page.locator('.product-title');
  //1)innerText() extracts plain text.eliminates white space and line breaks
  //textContent() extracts text including hidden elements.includes extra whitespace,line breaks,etc
  //console.log(await products.nth(1).innerText()); //14.1 inch laptop
  //console.log(await products.nth(1).textContent());
  /*const count=await products.count();
  for(let i=0;i<count;i++)
  {
    //const productname:string=await products.nth(i).innerText();//extracts plain text.eliminates white space and
    line breaks.
    //console.log(productname);
    //const productname:string | null=await products.nth(i).textContent();//extracts text including hidden
    elements.includes extra whitespace,line breaks,etc
    //console.log(productname);
    const productname:string | null=await products.nth(i).textContent();
    console.log(productname?.trim()); //trim to remove the extras.
  }*/
  //2)allInnerText()
  //allTextContent()
  console.log("Second methos Results:");
  //const productNames:string[]=await products.allInnerTexts();
  //console.log("Prodct name captured by allInnerText():",productNames);
  /*const productNames:string[]=await products.allTextContents();
  console.log("Prodct name captured by allTextContent():",productNames);
  const productNameTrimmed:string[]=productNames.map(text=>text.trim());
  console.log("Product name after trim:",productNameTrimmed);*/
  //3) all()
  const productslocator:Locator[]=await products.all();
  console.log(productslocator);
  //console.log(await productslocator[3].innerText());
  /*for(let productloc of productslocator)
  {
    console.log(await productloc.innerText());
  }*/
  //for in loop
  for(let i in productslocator)
  {
    console.log(await productslocator[i].innerText());
  }
});
```

9. File: csslocator.spec.ts

```
//CSS
//css is cascading style sheets
//css is used to style html elements
//css is used to make html elements look better
//css is used to change the layout of html elements
//css is used to make html elements responsive
//css is used to add animations to html elements
//css is used to add transitions to html elements
//css is used to add effects to html elements
//css is used to create themes for html elements
//css is used to create custom styles for html elements
//types of css locators
// 1) absolute locator
// 2) relative locator
// 3) attribute locator
/* tag with id tag#id or #id */
/* tag with class tag.class or .class */
/* tag with attribute tag[attribute='value'] or [attribute='value'] */
/* tag with multiple classes tag.class1.class2 or .class1.class2 */
/* tag with multiple attributes tag[attribute1='value1'][attribute2='value2'] or [attribute1='value1']
[attribute2='value2'] */
/* tag with pseudo-class tag:pseudo-class or :pseudo*/
// # represents id
// . represents class
import { expect,test } from '@playwright/test';
test("verify css locators",async( {page})=>{
    await page.goto("https://demowebshop.tricentis.com/");
    //tag with id
    await expect(page.locator("input#small-searchterms")).toBeVisible();
    await page.locator("input#small-searchterms").fill("T-Shirts");
    //tag with class
    await page.locator("input.search-box-text").fill("T-Shirts");
    //tag with attribute
    //tag[attribute='value']
    await page.locator(".search-box-text[value='Search store']").fill("T-Shirts");
});
//Absolute locator starts from the root (html) and follows the full DOM path to the element, making it brittle
and not recommended for automation.
//What “absolute locator” means
//It tells the browser/test tool:
//“Go from html → body → first div → next div → then p ... exactly in this order.”
```


10. File: dialogue.spec.ts

```
//handling dialogue
//alert, confirm, prompt
//by default dialogue is auto accepted, but we can change it to auto dismiss
//we can also listen to dialogue event and get the message and type of dialogue
import { test, expect } from '@playwright/test';
test('Handling Dialogues Test', async ({ page }) => {
  await page.goto('https://testautomationpractice.blogspot.com/');
  page.on('dialog', (dialog) => {
    console.log("Dialogue type is:", dialog.type()); //return type of dialogue
    expect(dialog.type()).toContain("alert"); //assert dialogue type
    console.log("Dialogue message is:", dialog.message()); //return message of dialogue
    expect(`${dialog.message()}`).toContain("I am an alert box!"); //assert dialogue message
    dialog.accept(); //auto accept dialogue
  });
  await page.locator("#alertBtn").click(); //open alert dialogue
  await page.waitForTimeout(5000);
});
test('Confirmation Dialogue Test', async ({ page }) => {
  await page.goto('https://testautomationpractice.blogspot.com/');
  //Register a dialog handler for confirm dialogue
  page.on('dialog', (dialog) => {
    console.log("Dialogue type is:", dialog.type()); //return type of dialogue
    expect(dialog.type()).toContain("confirm"); //assert dialogue type
    console.log("Dialogue text is:", dialog.message()); //return message of dialogue
    expect(dialog.message()).toContain("Press a button!"); //assert dialogue message
    dialog.dismiss(); //close the dialogue by clicking cancel
  });
  await page.locator("#confirmBtn").click(); //open confirm dialogue
  const text:string=await page.locator("#demo").innerText(); //get the text of element after closing the dialogue
  console.log("Text after closing the dialogue is:", text);
  await expect(page.locator("#demo")).toHaveText("You pressed Cancel!"); //assert the text after closing the dialogue
  await page.waitForTimeout(5000);
});
test.only('Prompt Dialogue Test', async ({ page }) => {
  await page.goto('https://testautomationpractice.blogspot.com/');
  page.on('dialog', (dialog) => {
    console.log("Dialogue type is:", dialog.type()); //return type of dialogue
    expect(dialog.type()).toContain("prompt"); //assert dialogue type
    console.log("Dialogue text is:", dialog.message()); //return message of dialogue
    expect(dialog.message()).toContain("Please enter your name:"); //assert dialogue message
    expect(dialog.defaultValue()).toContain("Harry Potter");
    dialog.accept('John'); //close the dialogue by providing a value
  });
  await page.locator("#promptBtn").click(); //open prompt dialogue
```

```
const text:string=await page.locator("#demo").innerText(); //get the text of element after closing the dialogue
console.log("Text after closing the dialogue is:",text);
await expect(page.locator("#demo")).toHaveText("Hello John! How are you today?"); //assert the text after
closing the dialogue
await page.waitForTimeout(5000);
});
```

11. File: downloadfile.spec.ts

```
//download files
//used to download files from the application under test
//since file download dialog box is a system level dialog box which cannot be handled by playwright, we can
use the page.waitForEvent('download') method to wait for the download event and then use the
download.saveAs() method to save the downloaded file to a specific location
import { test, expect } from '@playwright/test';
import fs from 'fs';
test('Text File Download', async ({ page }) => {
  await page.goto('https://testautomationpractice.blogspot.com/p/download-files_25.html');
  await page.locator("#inputText").fill("Hello World"); //provide some text to be downloaded
  await page.locator("#generateTxt").click(); //click on the generate text file button
  //start waiting for the download event before clicking on the download link
  const [download]=await Promise.all(
    [
      page.waitForEvent('download'), //wait for the download event
      page.locator("#txtDownloadLink").click() //click on the download text file button
    ]
  );
  //save the downloaded file to a specific location
  const downloadpath='downloads/HelloWorld.txt'; //provide the path where the downloaded file will be saved
  await download.saveAs(downloadpath); //save the downloaded file to the specified location
  //check if file exists in the specified location
  const fileexists=fs.existsSync(downloadpath); //check if the file exists in the specified location
  expect(fileexists).toBeTruthy(); //assert that the file exists in the specified location
  console.log("File downloaded successfully...."); //print the status message
  //cleanup - delete the downloaded file after the test is completed
  if(fileexists)
  {
    fs.unlinkSync(downloadpath); //delete the downloaded file
    console.log("Downloaded file deleted successfully...."); //print the status message
  }
  await page.waitForTimeout(5000);
});
//pdf file download
test.only('PDF File Download', async ({ page }) => {
  await page.goto('https://testautomationpractice.blogspot.com/p/download-files_25.html');
  await page.locator("#inputText").fill("Hello World"); //provide some text to be downloaded
  await page.locator("#generatePdf").click(); //click on the generate text file button

  //start waiting for the download event before clicking on the download link
  const [download]=await Promise.all(
    [
      page.waitForEvent('download'), //wait for the download event
      page.locator("#pdfDownloadLink").click() //click on the download text file button
    ]
  );
  //save the downloaded file to a specific location
```

```
const downloadpath='downloads/HelloWorld.pdf'; //provide the path where the downloaded file will be saved
await download.saveAs(downloadpath); //save the downloaded file to the specified location
//check if file exists in the specified location
const fileexists=fs.existsSync(downloadpath); //check if the file exists in the specified location
expect(fileexists).toBeTruthy(); //assert that the file exists in the specified location
console.log("File downloaded successfully...."); //print the status message
//cleanup - delete the downloaded file after the test is completed
if(fileexists)
{
    fs.unlinkSync(downloadpath); //delete the downloaded file
    console.log("Downloaded file deleted successfully...."); //print the status message
}
await page.waitForTimeout(2000);
});
```

12. File: dropdown.spec.ts

```
//Static Drop down list:
import {test,expect,Locator} from '@playwright/test';
test("Single select drop down",async( {page})=>{
  await page.goto('https://testautomationpractice.blogspot.com/');
  //select option from the drop down
  //by using visible text
  await page.locator('#country').selectOption('india');
  await page.waitForTimeout(2000);
  //by using value attribute
  await page.locator('#country').selectOption({value:'uk'});
  await page.waitForTimeout(2000);
  //by using label
  await page.locator('#country').selectOption({label:'India'});
  await page.waitForTimeout(2000);
  //by using index
  await page.locator('#country').selectOption({index:3});
  await page.waitForTimeout(3000);
  //check number of option in the dropdown(count)
  const dropdownoption:Locator=page.locator('#country>option');
  await expect(dropdownoption).toHaveCount(10);
  const count=await dropdownoption.count();
  console.log("Number of countries:",count);
  await page.waitForTimeout(2000)
  //check an option present in the drop down:
  const optiontext:string[]=(await dropdownoption.allTextContents()).map(text=>text.trim());
  console.log(optiontext);
  expect(optiontext).toContain('Japan');
  //printing option from the drop down
  for(const option of optiontext)
  {
    console.log(option);
  }
  await page.waitForTimeout(3000);
});
//Multi drop down
test("Multi select drop down",async( {page})=>{
  await page.goto('https://testautomationpractice.blogspot.com/');
  //select multiple option
  //using visible text
  await page.locator('#colors').selectOption(['Red','Blue','Green']);
  await page.waitForTimeout(3000);
  //using value attribute
  await page.locator('#colors').selectOption(['red','green','white']);
  await page.waitForTimeout(3000);
  //by using label
```

```

await page.locator('#colors').selectOption([
  { value: 'red'},
  { value: 'green'},
  { value: 'yellow'}
]);
await page.waitForTimeout(3000);
//using index
await page.locator('#colors').selectOption([ {index:0}, {index:1} ]);
await page.waitForTimeout(3000);
//check number of option in the dropdown(count)
const dropdownoption: Locator = page.locator('#colors > option');
await expect(dropdownoption).toHaveCount(7);
await page.waitForTimeout(3000);
//Check an option present in the drop down:
const optiontext: string[] = (await dropdownoption.allTextContents())
  .map(text => text.trim());
console.log(optiontext);
expect(optiontext).toContain('Green'); //check if the array contains green
await page.waitForTimeout(3000);
//printing option from the multi drop down
for(const option of optiontext)
{
  console.log(option);
}
await page.waitForTimeout(3000);
});

```

13. File: duplicatedropdown.spec.ts

```
//Duplicate drop down
import {test,expect,Locator} from '@playwright/test';
test("Single select drop down",async({page})=>{
  await page.goto('https://testautomationpractice.blogspot.com/');
  //Having duplicate
  const duplicateDropdown:Locator=page.locator('#colors>option');
  //not having duplicate
  //const duplicateDropdown:Locator=page.locator('#animals>option');//try animals
  const Dropdown:string[]=(await (duplicateDropdown.allTextContents())).map(text=>text.trim());
  //array and tuple can have duplicate value
  //Set cannot have duplicate value.
  const myset=new Set<string>(); //set
  const duplicates:string[]=[]; //array
  for(const text of Dropdown)
  {
    if(myset.has(text))
    {
      duplicates.push(text);
    }
    else{
      myset.add(text);
    }
  }
  console.log("Duplicated options are:",duplicates)
  if(duplicates.length>(0))
  {
    console.log("Duplicate option found:",duplicates);
  }
  else{
    console.log("No duplicate found..");
  }
  expect(duplicates.length).toBe(0);//2
});
```

14. File: dynamicelement.spec.ts

```
//The button text changes dynamically between START and STOP
//XPath uses logical OR:
//button[text()='STOP' or text()='START']
//Playwright automatically:
//Waits for the button
//Re-evaluates the XPath every time inside the loop
//using XPath to handle dynamic elements in Playwright
/*import { test, expect, Locator } from '@playwright/test';
test('Handle Dynamic Elements using XPath', async ({ page }) => {
  await page.goto('https://testautomationpractice.blogspot.com/');
  // Loop to click the button 5 times
  for (let i = 1; i <= 5; i++) {
    // XPath to handle dynamic START / STOP button text
    const button: Locator = page.locator(
      "//button[text()='STOP' or text()='START']"
    );
    // Click the button
    await button.click();
    // Wait for 2 seconds
    await page.waitForTimeout(2000);
  }
});*/
//using CSS to handle dynamic elements in Playwright
/*import { test, expect, Locator } from '@playwright/test';
test('Handle Dynamic Elements using CSS', async ({ page }) => {
  await page.goto('https://testautomationpractice.blogspot.com/');
  // Loop to click the button 5 times
  for (let i = 1; i <= 5; i++) {
    // CSS to handle dynamic START / STOP button text
    const button: Locator = page.locator(
      'button#start_stop'
    );
    // Click the button
    await button.click();
    // Wait for 2 seconds
    await page.waitForTimeout(2000);
  }
});*/
//using Playwright specific locators to handle dynamic elements
//getByRole with regex to match dynamic button text
import { test, expect } from '@playwright/test';
// Using Playwright specific locators
test('Handle Dynamic Elements using PW Locators', async ({ page }) => {
  await page.goto('https://testautomationpractice.blogspot.com/');
  // Loop to click the button 5 times
```



```
for (let i = 1; i <= 5; i++) {  
  // Locate button by role and dynamic name  
  const button = page.getByRole('button', { name: /START|STOP/ });  
  // Click the button  
  await button.click();  
  // Wait for 2 seconds  
  await page.waitForTimeout(2000);  
}  
});
```

15. File: dynamictable.spec.ts

```
//Dyanamic table
//means data keeps changing
import {test,expect,Locator} from "@playwright/test";
test("Dyanamic table",async( {page})=>{
  await page.goto("https://practice.expandtesting.com/dynamic-table");
  const table:Locator=page.locator("table.table tbody");
  await expect(table).toBeVisible();
  //select all the rows,then find number of rows
  const rows:Locator[]=await table.locator("tr").all();
  console.log("Number of rows in table:",rows.length);
  expect(rows).toHaveLength(4);
  //step1:for chrome process ges value of CPU load
  //Read each row 0 check chrome presence.
  let cpuLoad="";
  for(const row of rows)
  {
    const processrow:string=await row.locator("td").nth(0).innerText();
    if(processrow==="Chrome")
    {
      //cpuLoad=await row.locator('td:has-text("%)").innerText();
      cpuLoad=await row.locator('td:has-text("%)").innerText();
      console.log("CPU load of Chrome:",cpuLoad);
      break;
    }
  }
  //step 2: compare it with value in the yellow label.
  let yellowboxtext:string=await page.locator("#chrome-cpu").innerText();
  console.log("Chrome CPU load from yellow box:",yellowboxtext);
  if(yellowboxtext.includes(cpuLoad))
  {
    console.log("CPU load of chrome is equal.");
  }
  else{
    console.log("CPU load of chrome is not equal.");
  }
  expect(yellowboxtext).toContain(cpuLoad);
  await page.waitForTimeout(5000);
});
```

16. File: example.spec.ts

```
import { test, expect } from '@playwright/test';

test('has title', async ({ page }) => {
  await page.goto('https://playwright.dev/');
  // Expect a title "to contain" a substring.
  await expect(page).toHaveTitle(/Playwright/);
});

test('get started link', async ({ page }) => {
  await page.goto('https://playwright.dev/');
  // Click the get started link.
  await page.getByRole('link', { name: 'Get started' }).click();
  // Expects page to have a heading with the name of Installation.
  await expect(page.getByRole('heading', { name: 'Installation' })).toBeVisible();
});
```

17. File: frame.spec.ts

```
//frame
//frame is a class that represents a frame in the game
//it has a constructor that takes in the frame number and the rolls in the frame
//it has a method that calculates the score of the frame based on the rolls and the next rolls in the game
import { test, expect, Locator } from '@playwright/test';
import { url } from 'node:inspector';
test('Frames demo', async ({ page }) => {
  await page.goto('https://ui.vision/demo/webtest/frames/');
  // total frames in the page
  const frames=page.frames();
  console.log("Total frames in the page:",frames.length);
  /*//using frame locator to access the frame and perform actions
  const frame = page.frame({ url: /frame_1/ });
  if(frame){
    await frame.locator("[name='mytext1']").fill("Hello");
    //await frame.fill("[name='mytext1']", "Hello World");
  }
  else{
    console.log("Frame is not visible");
  }
  await page.waitForTimeout(5000);*/
  //using frame locator to access the frame and perform actions
  const frame=page.frameLocator("[src='frame_1.html']").locator("[name='mytext1']").fill("Hello World");
  await page.waitForTimeout(5000);
});
test.only('inner child frames demo', async ({ page }) => {
  await page.goto('https://ui.vision/demo/webtest/frames/');
  // Get frame_3
  const frame3 = page.frame({ url: /frame_3/ });
  if (frame3) {
    // Fill textbox in frame_3
    await frame3.locator("[name='mytext3']").fill("Hello World");
    // Wait until child frame loads
    await page.waitForTimeout(2000);
    const childFrames = frame3.childFrames();
    console.log("Child frames count:", childFrames.length);
    if (childFrames.length > 0) {
      const child = childFrames[0];
      // Use exact label text (Google form is case sensitive)
      const radio = child.getByLabel("I am a human");
      await radio.check();
      await expect(radio).toBeChecked();
    } else {
      console.log("Child frame not found");
    }
  }
}
```

```
} else {  
  console.log("Frame is not visible");  
}  
await page.waitForTimeout(5000);  
});
```

18. File: hiddendropdown.spec.ts

```
//Auto suggest drop down;
import { test, expect, Locator } from '@playwright/test';
test("Bootstrap hidden drop down",async( {page})=>{
await page.goto("https://opensource-demo.orangehrmlive.com/web/index.php/auth/login");
//Login page:
await page.locator('input[name="username"]').fill('Admin');
await page.locator('input[name="password"]').fill('admin123');
await page.locator('button[type="submit"]').click();

//click on the PIM:
await page.getByText('PIM').click();
// //click on job title:
await page.locator('form i').nth(2).click();
await page.waitForTimeout(3000);
//capture all the option from dropdown:
const options:Locator=page.locator("div[role='listbox'] span");
const count:number=await options.count();
console.log("Number of options in dropdown:",count);
//print all the options
console.log("All the text content:",await options.allTextContents());
console.log("printing all the options....");
for(let i=0;i<count;i++)
{
    //console.log(await options.nth(i).innerText());
    console.log(await options.nth(i).textContent());
}
//selecct/click on option
for(let i=0;i<count;i++)
{
    const text=await options.nth(i).innerText();
    if(text==='Automation Tester')
    {
        await options.nth(i).click();
        break;
    }
}
await page.waitForTimeout(5000);
});
```

19. File: html-runner.spec.ts

```
import { test, expect } from '@playwright/test';

test('Run local hosted HTML in Playwright', async ({ page }) => {
  await page.goto('http://127.0.0.1:5500/tests/testhtml.html');
  // getByRole (heading)
  await expect(
    page.getByRole('heading', { name: 'Playwright Locator Practice App' })
  ).toBeVisible();
  // getByText
  await expect(
    page.getByText('Practice all Playwright locators on one page')
  ).toBeVisible();
  // getByAltText
  await expect(
    page.getByAltText('Playwright Company Logo')
  ).toBeVisible();
  // getByLabel
  await page.getByLabel('First name:').fill('John');
  await page.getByLabel('Last name:').fill('Wick');
  await page.getByLabel('Email:').fill('john@test.com');
  // getByPlaceholder
  await page.getByPlaceholder('Enter first name').fill('Johnny');
  // getByRole (button)
  await page.getByRole('button', { name: 'Create Account' }).click();
  // getByTitle
  await expect(
    page.getByTitle('Click here to get help')
  ).toBeVisible();
  // getByTestId
  await expect(
    page.getByTestId('dashboard-card')
  ).toBeVisible();
});
```

20. File: infinitescroll.spec.ts

```
//infinte scroll test
// it is used to handle the infinite scrolling in the browser when the element is not visible in the viewport and we
need to scroll down to load more elements
import { test, expect } from '@playwright/test';
test('Infinite scroll demo', async ({ page }) => {
  test.slow(); // slow down the test execution to see the scrolling effect
  await page.goto('https://www.booksbykilo.in/new-books?pricerange=201to500');
  //window.scrollTo(0, document.body.scrollHeight); // scroll to the bottom of the page to load more elements
  let previousheight = 0;
  while(true)
  {
    await page.evaluate(() => {
      window.scrollBy(0, window.innerHeight); // scroll to the bottom of the page
      // wait for 2 seconds to load more elements

    });
    await page.waitForTimeout(2000);
    //capture the current height of the page after scrolling
    const currentheight =await page.evaluate(() => {
      return document.body.scrollHeight; // scroll to the bottom of the page to load more elements
    })
    console.log("Previous height: ",previousheight);
    console.log("Current height: ",currentheight);
    if(currentheight === previousheight)
    {
      break;
    }
    previousheight = currentheight;
  }
  console.log("Reached the end of the page");

});
```


21. File: infinitemscrollfindbook.spec.ts

```
import { test, expect } from '@playwright/test';

test('To find a book in infinite scroll demo', async ({ page }) => {
  test.slow(); // slow down the test execution to see the scrolling effect
  await page.goto('https://www.booksbykilo.in/new-books?pricerange=201to500');
  //window.scrollTo(0, document.body.scrollHeight); // scroll to the bottom of the page to load more elements
  let previousheight = 0;
  let bookFound = false;
  while(true)
  {
    const title = await page.locator('#productsDiv h3').allTextContents();
    if(title.includes("The Ink Black Heart"))
    {
      console.log("Book found");
      bookFound = true;
      expect(bookFound).toBeTruthy();
      break;
    }
    await page.evaluate(() => {
      window.scrollBy(0, window.innerHeight); // scroll to the bottom of the page
      // wait for 2 seconds to load more elements

    });
    await page.waitForTimeout(2000);
    //capture the current height of the page after scrolling
    const currentheight =await page.evaluate(() => {
      return document.body.scrollHeight; // scroll to the bottom of the page to load more elements
    })
    console.log("Previous height: ",previousheight);
    console.log("Current height: ",currentheight);
    if(currentheight === previousheight)
    {
      break;
    }
    previousheight = currentheight;
  }
  console.log("Reached the end of the page");
  if(!bookFound){
    console.log("Book not found");
  }
});
```

22. File: keyboardaction.spec.ts

```
//Keyboard Action
//it is used to perform keyboard actions like key press, key down, key up etc.
//keyboard methods:
//keyboard methods are used to perform keyboard actions like key press, key down, key up etc.
//keyboard.press(key, options) - it is used to press a key.
//keyboard.down(key, options) - it is used to press a key down.
//keyboard.up(key, options) - it is used to release a key.
//insertText,down,press,type,up are the methods of keyboard class which are used to perform keyboard actions.
import { test, expect } from '@playwright/test';
test('Keyboard Action', async ({ page }) => {
  await page.goto('https://testautomationpractice.blogspot.com/');
  const input1=page.locator('#input1');
  //1.focus on the input field
  await input1.focus(); //focus|click
  //2.provide some text in the input field
  await page.keyboard.insertText('Hello World'); //type|insertText
  //3.ctrl+a to select all the text
  await page.keyboard.down('Control');
  await page.keyboard.press('a');
  await page.keyboard.up('Control');
  //4.ctrl+c to copy the text
  await page.keyboard.down('Control');
  await page.keyboard.press('c');
  await page.keyboard.up('Control');
  //5.press tab twice to move to the next input field
  await page.keyboard.press('Tab');
  await page.keyboard.press('Tab');
  //6.ctrl+v to paste the text in the next input field
  await page.keyboard.down('Control');
  await page.keyboard.press('v');
  await page.keyboard.up('Control');
  //7.press tab twice to move to the next input field
  await page.keyboard.press('Tab');
  await page.keyboard.press('Tab');
  //8.ctrl+v to paste the text in the next input field
  await page.keyboard.down('Control');
  await page.keyboard.press('v');
  await page.keyboard.up('Control');
  await page.waitForTimeout(5000);
});
test.only('Keyboard Action -simple mode', async ({ page }) => {
  await page.goto('https://testautomationpractice.blogspot.com/');
  const input1=page.locator('#input1');
  //1.focus on the input field
  await input1.focus(); //focus|click
```

```
//2.provide some text in the input field
  await page.keyboard.insertText('Hello World'); //type|insertText
//3.ctrl+a to select all the text
  await page.keyboard.press('Control+a'); //shortcut for select all
//4.ctrl+c to copy the text
  await page.keyboard.press('Control+c'); //shortcut for copy
//5.press tab twice to move to the next input field
  await page.keyboard.press('Tab');
  await page.keyboard.press('Tab');
//6.ctrl+v to paste the text in the next input field
  await page.keyboard.press('Control+v'); //shortcut for paste
//7.press tab twice to move to the next input field
  await page.keyboard.press('Tab');
  await page.keyboard.press('Tab');
//8.ctrl+v to paste the text in the next input field
  await page.keyboard.press('Control+v'); //shortcut for paste
  await page.waitForTimeout(5000);
});
```

23. File: locators.spec.ts

//Locators are the central piece of Playwright's auto-waiting and retry-ability. In a nutshell, locators represent a way to find element(s) on the page at any moment.

//Locators are lazy: they do not perform any action or query until an action is performed on them. This allows locators to always point to the most up-to-date element matching the selector, even if the DOM changes between the time the locator is created and the time an action is performed on it.

//Locators are auto-waiting: Playwright automatically waits for the element to be ready before performing actions on it. This includes waiting for the element to be attached to the DOM, visible, stable, and enabled.

//Locators are retry-able: If an action fails because the element is not ready, Playwright will retry the action until it succeeds or a timeout is reached.

/*page.getByRole() to locate by explicit and implicit accessibility attributes.

page.getByText() to locate by text content.

page.getByLabel() to locate a form control by associated label's text.

page.getByPlaceholder() to locate an input by placeholder.

page.getByAltText() to locate an element, usually image, by its text alternative.

page.getByTitle() to locate an element by its title attribute.

page.getByTestId() to locate an element based on its data-testid attribute (other attributes can be configured).*/

```
import {expect, Locator, test} from '@playwright/test';
```

```
test("Verify locators on the page", async ({ page }) => {
```

```
  await page.goto("https://demo.nopcommerce.com/");
```

```
  // page.getByAltText() to locate an element, usually image, by its text alternative.
```

```
  const logo: Locator = page.getByAltText("nopCommerce demo store");
```

```
  await expect(logo).toBeVisible();
```

```
  //await logo.click();
```

```
  //page.getByText() to locate by text content.
```

```
  //used for static text on the page like headings, paragraphs, links, etc.
```

```
  //const text:Locator= page.getByText("Welcome to our store");
```

```
  //await expect(text).toBeVisible();
```

```
  await expect(
    page.getByText("Welcome to our store")
  ).toBeVisible();
```

```
  //page.getByRole() to locate by explicit and implicit accessibility attributes.
```

```
  //role includes button, link, checkbox, textbox, heading, etc.
```

```
  await Promise.all([
    page.waitForURL(/\/register/),
    page.getByRole("link", { name: "Register" }).click(),
  ]);
```

```
  // □ Reliable assertion: URL (not blocked by Cloudflare UI)
```

```
  await expect(page).toHaveURL(/\/register/);
```

```
  //page.getByLabel() to locate a form control by associated label's text.
```

```
  //used for input fields, text areas, dropdowns etc.
```

```
  await expect(page.getByLabel("First name:")).toBeVisible();
```

```
  await page.getByLabel("First name:").fill("John");
```

```
  await page.getByLabel("Last name:").fill("Wick");
```

```
  await page.getByLabel("Email:").fill("johnwick@gmail.com");
```

```
  //page.getByPlaceholder() to locate an input by placeholder.
```

```
//best for input fields where placeholder text is descriptive  
await page.getByPlaceholder("Search Store").fill("laptop");  
});
```

24. File: mouseaction.spec.ts

```
//Mouse action tests
// it is used to handle the mouse actions in the browser by using the mouse object
import { test, expect } from '@playwright/test';
test('Mouse action demo', async ({ page }) => {
  await page.goto('https://testautomationpractice.blogspot.com/');
  const pointme=page.locator('.dropbtn');
  await pointme.hover();
  const laptops=page.locator('.dropdown-content a:nth-child(2)');
  laptops.hover();
  await page.waitForTimeout(5000);
});
test('Mouse Right click action demo', async ({ page }) => {
  await page.goto('https://swisnl.github.io/jQuery-contextMenu/demo.html');
  const button=page.locator('span.context-menu-one');
  await button.click({button:'right'}); // right click action
  await page.waitForTimeout(3000);
});
test('Double click action demo', async ({ page }) => {
  await page.goto('https://testautomationpractice.blogspot.com/');
  const btncopy=page.locator('button[ondblclick="myFunction1()"]');
  await btncopy.dblclick(); // double click action
  const textfield=page.locator('#field2');
  await expect(textfield).toHaveValue('Hello World!'); // assertion to verify the double click action
  await page.waitForTimeout(3000);
});
test.only('Drag and Drop action demo', async ({ page }) => {
  await page.goto('https://codepen.io/EpsilonDeltaCriterion/pen/jLoPgE');
  // Target ONLY the result iframe
  const frame = page.frameLocator('iframe.result-iframe');
  const rome = frame.locator('#box6');
  const italy = frame.locator('#box106');
  //await rome.dragTo(italy);
  await rome.hover();
  await page.mouse.down();
  await italy.hover();
  await page.mouse.up();
  //approach 2: using mouse move and mouse down and mouse up actions
  const washington = frame.locator('#box3');
  const usa = frame.locator('#box103');
  await washington.dragTo(usa); // using dragTo method
  await page.waitForTimeout(5000);
});
```

25. File: mytest.spec.ts

```
//syntax
/*test( "title",()=> {
  //step 1
  //step 2
  //assertion
});*/
/*test("Verify the title of the page",async ({page})=> {
  //step 1: navigate to the url
  await page.goto("https://demo.opencart.com/");
  let title:string = await page.title();
  console.log("title of the page is: ", title);
  await expect(page).toHaveTitle("Your Store");
}); */
```

//await is used to pause code execution until an asynchronous operation finishes, so the next line runs with valid data and state.

```
import {expect, test} from '@playwright/test';
test("Verify the title of the page", async ({ page }) => {
  await page.goto(
    "https://gauravkhurana.in/test-automation-play/",
    { waitUntil: "domcontentloaded" }
  );
  await expect(page).toHaveTitle("Test Automation Practice Hub");
  const title = await page.title();
  console.log("Title of the page is:", title);
  const url = page.url();
  console.log("Current URL is:", url);
  await expect(page).toHaveURL(
    "https://gauravkhurana.in/test-automation-play/"
  );
});
```

26. File: mytest2.spec.ts

```
import { test, expect } from '@playwright/test';
test("Verify the title of the page", async ({ page }) => {
  await page.goto("https://testautomationpractice.blogspot.com/");
  await expect(page).toHaveTitle(/Test Automation Practice Hub/);
  console.log("Title:", await page.title());
  console.log("URL:", page.url());
  await expect(page).toHaveURL(/testautomationpractice\.blogspot\.com/);
});
```


27. File: paginationtable.spec.ts

```
//paginationtable
//A pagination table is a table that splits large data into multiple pages instead of showing everything at once.
import { test, expect, Locator } from "@playwright/test";
test("paginationtable", async ({ page }) => {
  await page.goto("https://datatables.net/examples/basic_init/zero_configuration.html");
  let hasmorepages=true;
  //const rows=await page.locator("#example tbody tr").all();
  /*for(let row of rows)
  {
    console.log(await row.innerText());
  }*/
  while(hasmorepages)
  {
    const rows=await page.locator("#example tbody tr").all();
    for(const row of rows)
    {
      console.log(await row.innerText());
    }
  }
  await page.waitForTimeout(2000);
  //button[aria-label='Next']
  //button[aria-controls='example']:has-text(",")
  //button[aria-controls='example']:nth-child(9 )
  const nextButton:Locator=page.locator("button[aria-label='Next']");
  const isDisabled=await nextButton.getAttribute('class');
  if(isDisabled?.includes("disabled"))
  {
    hasmorepages=false;
  }
  else{
    await nextButton.click();
  }
}
//await page.waitForTimeout(5000);
});
test("Filter rows and check the rows count", async ({ page }) => {
  await page.goto("https://datatables.net/examples/basic_init/zero_configuration.html");
  const dropdowns:Locator=await page.locator('#dt-length-0');
  await dropdowns.selectOption({label:'25'});
  //Approach 1
  const rows=await page.locator("#example tbody tr").all();
  expect(rows.length).toBe(25);//assertion
  //Approach 2
  const rows2=await page.locator("#example tbody tr");
  await expect(rows2).toHaveCount(25);
});
```

```
test.only("Searching rows and check the rows count", async ({ page }) => {
  await page.goto("https://datatables.net/examples/basic_init/zero_configuration.html");
  const searchbox:Locator=page.locator('#dt-search-0');
  await searchbox.fill('Paul Byrd');
  await page.waitForTimeout(5000);
  const rows=await page.locator("#example tbody tr").all();
  if(rows.length>=1)
  {
    let matchfound=false;
    for(let row of rows)
    {
      const text=await row.innerText();
      if(text.includes('Paul Byrd'))
      {
        console.log("Record exist- found");
        matchfound=true;
        break;
      }
    }
    //expect(matchfound).toBe(true);
    expect(matchfound).toBeTruthy();
  }
  else{
    console.log("Record not found");
  }
});
```

28. File: popups.spec.ts

```
//popups
//popups is a library for creating popups in the browser. It is built on top of the native browser APIs and
//provides a simple and consistent interface for creating and managing popups.
import { test, expect, Locator, Page, chromium } from '@playwright/test';
test('Popups demo', async ({browser}) => {
  const context=await browser.newContext();
  const page=await context.newPage();
  await page.goto('https://testautomationpractice.blogspot.com/');
  //multiple popups
  //page.waitForEvent('popup');
  //await page.locator('#PopUp').click();
  await Promise.all([page.waitForEvent('popup'), page.locator('#PopUp').click()]);
  const allpopupwindows=context.pages();
  console.log("Number of popups opened:",allpopupwindows.length);
  console.log(allpopupwindows[0].url());
  console.log(allpopupwindows[1].url());
  for(const pw of allpopupwindows)
  {
    const title=await pw.title();
    if(title.includes("Playwright"))
    {
      await pw.locator('.getStarted_Sjson').click();
      await pw.close();
    }
  }
  await page.waitForTimeout(5000);
});
```

29. File: sorted.spec.ts

```
//sorted dropdown
import {test,expect,Locator} from '@playwright/test';
test("Single select drop down",async({page})=>{
  await page.goto("https://testautomationpractice.blogspot.com/");
  //const Dropdownoption:Locator=page.locator('#colors>option');
  const Dropdownoption:Locator=page.locator('#animals>option');//try animals
  console.log(await Dropdownoption.allTextContents());
  const optiontext:string[]=(await Dropdownoption.allTextContents()).map(text=>text.trim());
  const originallist:string[]=[...optiontext];//... spread operator
  const sortedlist:string[]=[...optiontext.sort()]; //array sort is mutable
  console.log("original list",originallist);
  console.log("sorted list",sortedlist);
  expect(originallist).toEqual(sortedlist); //this will give error
  //await page.waitForTimeout(3000);
});
```

30. File: statictable.spec.ts

```
import {expect, Locator, test} from "@playwright/test";
test("Static Table", async ({page}) => {
  await page.goto("https://testautomationpractice.blogspot.com/");
  const table: Locator = page.locator("table[name='BookTable'] tbody");
  await expect(table).toBeVisible();
  //1) count number of rows in a table:
  //const rows: Locator = page.locator("table[name='BookTable'] tbody tr"); // returns all the rows including header
  const rows: Locator = table.locator("tr");
  await expect(rows).toHaveCount(7); //approach 1
  const rowcounts: number = await rows.count();
  console.log("Number of rows in a table", rowcounts);
  expect(rowcounts).toBe(7); //approach 2
  //2) Count number of headers/column
  //const columns: Locator = page.locator("table[name='BookTable'] tbody tr th");
  const columns: Locator = rows.locator("th");
  await expect(columns).toHaveCount(4);
  const columncounts: number = await columns.count();
  console.log("Number of columns in a table", columncounts);
  expect(columncounts).toBe(4);
  //3) Read all data from 2nd row (index 2 means 3rd row including header)
  const secondrowcells: Locator = rows.nth(2).locator("td");
  const secondrowText: string[] = await secondrowcells.allInnerTexts();
  console.log("2nd row data:", secondrowText); //2nd row data: [ 'Learn Java', 'Mukesh', 'Java', '500' ]
  await expect(secondrowcells).toHaveText([ 'Learn Java', 'Mukesh', 'Java', '500' ]); //assertion
  console.log("Printing 2nd row data.....");
  for (let text of secondrowText)
  {
    console.log(text);
  }
  //4) Read all data from the table (Excluding header)
  console.log("Printing all data from the table.....");
  const allrowdata = await rows.all();
  console.log("BookName\tAuthor\tSubject\tPrice");
  for (let row of allrowdata.slice(1)) //slice skip header row
  {
    const cols = await row.locator('td').allInnerTexts();
    console.log(cols.join('\t'));
  }
  //5) print book names where author is Mukesh
  console.log("Printing book names where author is Mukesh.....");
  const mukeshbook: string[] = [];
  for (let row of allrowdata.slice(1))
  {
    const cells = await row.locator('td').allInnerTexts();
    const author = cells[1];
```

```
const BookName=cells[0];
if(author==='Mukesh')
{
    console.log(`${author}\t${BookName}`);
    mukeshbook.push(BookName);

}
}
expect(mukeshbook).toHaveLength(2);//ASSERTION
//Calculate total price of all book
let totalprice:number=0;
console.log("Printing total price of all book.....");
for(let row of allrowdata.slice(1))
{
    const cells=await row.locator('td').allInnerTexts();
    const price=cells[3];
    totalprice=totalprice+parseInt(price);
}
console.log("Total price:",totalprice);
expect(totalprice).toBe(7100);//assertion
});
```

31. File: tabs.spec.ts

```
//tabs
//tabs should be able to switch between different content when clicked
//tabs is a component that allows users to switch between different content by clicking on the tab headers
import {test,expect,chromium,Locator} from '@playwright/test';
test('Tabs demo', async () => {
  const browser=await chromium.launch();
  const context=await browser.newContext();
  //creating 1 page
  const parentpage=await context.newPage();
  parentpage.goto('https://testautomationpractice.blogspot.com/');
  // 2 statements should be executed at the same time, one is waiting for the new page to open and the other is
  clicking on the button that opens the new page
  //context.waitForEvent('page'); //pending,fullfilled,rejected
  //parentpage.locator("button:has-text('New tab')").click(); //opens new tab6
  const [childpage]=await Promise.all([context.waitForEvent('page'),parentpage.locator("button:has-text('New
  tab')").click()]);
  //approach 1:switch between pages and get the title of the page
  const pages=context.pages();
  console.log("Number of pages created:",pages.length);
  console.log("Title of the parent page:",await pages[0].title());
  console.log("Title of the child page:",await pages[1].title());
  //approach 2:alternative way to switch between pages and get the title of the page
  console.log("Title of the parent page:",await parentpage.title());
  console.log("Title of the child page:",await childpage.title());
  await parentpage.waitForTimeout(5000);
  await childpage.waitForTimeout(5000);
});
```

32. File: uploadfile.spec.ts

```
//upload files
//used to upload files to the application under test
//filechooser is a class which is used to handle file upload dialog box
//setInputFiles is a method of filechooser class which is used to set the file to be uploaded
//since file upload dialog box is a system level dialog box which cannot be handled by playwright, we can use
setInputFiles method to set the file to be uploaded
import { test, expect } from '@playwright/test';
test('File Upload', async ({ page }) => {
  await page.goto('https://testautomationpractice.blogspot.com/');
  await page.locator("#singleFileInput").setInputFiles('upload/Day26-Tables-Lab+Assignment.pdf'); //provide
the path of the file to be uploaded
  await page.locator("button:has-text('Upload single file ')").click(); //click on the upload button
  const msg=await page.locator("#singleFileStatus").textContent(); //get the text of the status message
  expect(msg).toContain('Day26-Tables-Lab+Assignment.pdf'); //assert the status message
  console.log("Upload successful...."); //print the status message
  await page.waitForTimeout(5000);
});
//multiple file upload
test.only('Multiple File Upload', async ({ page }) => {
  await page.goto('https://testautomationpractice.blogspot.com/');
  await page.locator("#multipleFilesInput").setInputFiles(['upload/Day26-Tables-Lab+Assignment.pdf','upload/
Day27-DatePickers-Labs.pdf']); //provide the path of the files to be uploaded
  await page.locator("button:has-text('Upload multiple files')").click(); //click on the upload button
  const msg=await page.locator("#multipleFilesStatus").textContent(); //get the text of the status message
  expect(msg).toContain('Day26-Tables-Lab+Assignment.pdf'); //assert the status message
  expect(msg).toContain('Day27-DatePickers-Labs.pdf'); //assert the status message
  console.log("Multiple Upload successful...."); //print the status message
  await page.waitForTimeout(5000);
});
```


33. File: xpath.spec.ts

```
//xpath is syntax used to navigate through elements and attributes in an XML document.
//xpath is used to locate elements in a web page for automation testing
//xpath is used in selenium to locate elements
//xpath is used in playwright to locate elements
//xpath is used in many other automation tools to locate elements
//types of xpath
//1) absolute xpath
//2) relative xpath
//absolute xpath starts from the root element and follows the full path to the target element
//single slash (/) is used in absolute xpath
//like html/body/div/div/p
//relative xpath starts from anywhere in the document and uses double slash (//)
//double slash (//) is used in relative xpath
//like //tagName[@attribute='value']
//like //tagName[text()='textvalue']
//And operator in xpath
//used to combine multiple conditions in a single xpath expression
//syntax: //tagName[@attribute1='value1' and @attribute2='value2']
//example:
//input[@type='submit' and @value='Search']
//Or operator in xpath
//used to specify multiple conditions where at least one must be true
//syntax: //tagName[@attribute1='value1' or @attribute2='value2']
//example:
//input[@type='submit' or @value='Search']
import { expect, test, Locator } from '@playwright/test';
import { count } from 'node:console';
test("XPath demo in Playwright", async ({ page }) => {
  await page.goto("https://demowebshop.tricentis.com/");
//absolute xpath
//use xpath or double slash(//) because it take single slash as css xpath= or //html
const absolutelogo:Locator=page.locator("xpath=/html[1]/body[1]/div[4]/div[1]/div[1]/div[1]/a[1]/img[1]");
await expect(absolutelogo).toBeVisible();
//relative xpath
const relativelogo:Locator=page.locator("//img[@alt='Tricentis Demo Web Shop']");
await expect(relativelogo).toBeVisible();
//contains() function in xpath
//used to find elements that contain a specific text or attribute value
//syntax: //tagName[contains(@attribute,'value')]
const products :Locator=page.locator("//h2//a[contains(@href,'computer')]");
await products.first().isVisible();
const productCount:number=await products.count();
console.log("Number of products found with 'computer' in href: "+productCount);
expect(productCount).toBeGreaterThan(0);
//textContent used to get the text inside the element
```

```

//get text from ALL matching elements
const productNames = await products.allTextContents();
console.log(productNames);
//get text from specific element
//by using first(), last(), nth() methods
console.log("First computer related product: ",await products.first().textContent());
console.log("Last computer related product: ",await products.last().textContent());
console.log("Nth computer related product: ",await products.nth(1).textContent());
//by using array index
console.log("first computer related product: ",await products.nth(0).textContent());
console.log("second computer related product: ",await products.nth(1).textContent());
//starts-with() function in xpath
//used to find elements whose attribute value starts with a specific string
//syntax: //tagname[starts-with(@attribute,'value')]
const searchBox:Locator=page.locator("//h2/a[starts-with(@href,'build')]");
const count: number = await searchBox.count(); // □ CALL + await
expect(count).toBeGreaterThan(0);
//text() function in xpath
//used to find elements based on their exact text content
//syntax: //tagname[text()='textvalue']
const reglink:Locator=page.locator("//a[text()='Register']");
await expect(reglink).toBeVisible();
//last() function in xpath
//used to select the last element in a set of matching elements
//syntax: (//tagname)[last()]
const lastProduct:Locator=page.locator("//div[@class='column follow-us']/li[last()]");
console.log("Last product on the page: ",await lastProduct.textContent());
await expect(lastProduct).toBeVisible();
console.log("Text content of last product: ",await lastProduct.textContent());
//position() function in xpath
//used to select an element based on its position in a set of matching elements
//syntax: (//tagname)[position()=n]
const positionitem:Locator=page.locator("//div[@class='column follow-us']/li[position()=3]");
console.log("Third product on the page: ",await positionitem.textContent());
await expect(positionitem).toBeVisible();
});

```

34. File: xpathaxes.spec.ts

```
//xpath axes
//defines the relationship between the current node and nodes in the document.
//allow you to navigate through the XML or HTML document structure in a flexible way.
//parent
//parent axis selects the parent node of the current node.
//example: /bookstore/book/parent::bookstore
//child
//child axis selects all child nodes of the current node.
//example: /bookstore/child::book
//ancestor
//ancestor axis selects all ancestor nodes (parent, grandparent, etc.) of the current node.
//example: /bookstore/book/ancestor::library
//descendant
//descendant axis selects all descendant nodes (children, grandchildren, etc.) of the current node.
//example: /bookstore/descendant::book
//following
//following axis selects all nodes that come after the current node in the document, excluding descendants.
//example: /bookstore/book/following::author
//following-sibling
//following-sibling axis selects all sibling nodes that come after the current node.
//example: /bookstore/book/following-sibling::book
//preceding
//preceding axis selects all nodes that come before the current node in the document, excluding ancestors.
//example: /bookstore/book/preceding::magazine
//preceding-sibling
//preceding-sibling axis selects all sibling nodes that come before the current node.
//example: /bookstore/book/preceding-sibling::book
//self
//self axis selects the current node itself.
//example: /bookstore/book/self::book
import {test,expect,Locator} from '@playwright/test';
test("xpath axes demo",async ({page})=>{
    await page.goto("https://www.w3schools.com/html/html_tables.asp");
    //self axis
    const germanycell:Locator=page.locator("//td[text()='Germany']/self::td");
    await expect(germanycell).toHaveText("Germany");
    //parent axis
    const parentRow:Locator=page.locator("//td[text()='Germany']/parent::tr");
    await expect(parentRow).toContainText("Maria Anders");
    console.log(await parentRow.textContent());
    //child axis
    const secondRowChildren:Locator=page.locator("//table[@id='customers']/child::tbody/child::tr[2]/child::td");
    await expect(secondRowChildren).toHaveLength(3);
    //ancestor axis
    const ancestorTable:Locator=page.locator("//td[text()='Germany']/ancestor::table");
```

```
await expect(ancestorTable).toHaveAttribute("id","customers");
//descendant axis
const tableDescendants:Locator=page.locator("//table[@id='customers']/descendant::td");
await expect(tableDescendants).toHaveCount(18); //try 15
//following axis
const followingNodes:Locator=page.locator("(//td[normalize-space()='Germany'])[1]/following::td");
await expect(followingNodes).toHaveText("Centro comercial Moctezuma"); //try 12
//following-sibling axis
const followingSiblings:Locator=page.locator("//td[text()='Germany']/following-sibling::td");
await expect(followingSiblings).toHaveText("UK");
//preceding axis
const precedingNodes:Locator=page.locator("(//td[normalize-space()='Germany'])[1]/preceding::td");
await expect(precedingNodes).toHaveText("Maria Anders"); //try 6
});
```